

Pipeline Provenance & Regeneration Audit Report

Date: 2026-03-21

Scope: All figures/tables referenced in `README.md` , `docs/core/*.md` , `docs/analysis/*.md` (excluding `docs/archive/`)

Status: Audit only — NO files overwritten, NO push, NO PR

=== 1. PROVENANCE TABLE ===

1A. Figures Embedded in Technical Report (docs/core/technical_report.md)

#	Artifact	Location	Generator	Upstream Data	Status
TR-1	policy_comparison_panel_K20_cap2.png	results/baseline/figures/	scripts/run_producti on_v2.py	Optimization solver + simulation	✅ Reproducible
TR-2	re-sponse_time_distribution_by_policy.png	results/baseline/figures/	scripts/run_producti on_v2.py	all_results_raw.csv	✅ Reproducible
TR-3	fleet_sensitivity_dual.png	results/archive/figures/	scripts/analysis/regenerate_all_figures.py	all_results_raw.csv	⚠️ Archive artifact in active doc
TR-4	cbd_robustness_enhanced.png	results/analysis/figures/	scripts/analysis/regenerate_all_figures.py	cbd_experiment_results.csv	⚠️ Generator has stale read path
TR-5	queue_comparison_by_policy.png	results/analysis/figures/	scripts/analysis/analyze_queue_metrics.py	Production exp CSVs	⚠️ Generator has stale read path
TR-6	seasonal_patterns.png	results/analysis/figures/	scripts/analysis/analyze_seasonal_patterns.py	crashes_manhattan.csv	⚠️ Generator writes to stale output path
TR-7	distance_comparison_bar.png	results/analysis/distance_comparison/	scripts/analysis/run_distance_comparison_experiment.py	Distance matrices + simulation	✅ Reproducible
TR-8	cbd_equity_tradeoff_summary.png	results/analysis/figures/	scripts/analysis/regenerate_all_figures.py	cbd_experiment_results.csv	⚠️ Generator has stale read path
TR-9	capacity_sensitivity		scripts/analysis/gener-		🔴 Archive + generator

#	Artifact	Location	Generator	Upstream Data	Status
	vity_heatmap.png	results/archive/figures/	ate_capacity_sensitivity_heatmap.py	simulation_results.csv	broken (stale read path) + shows “Data format issue”

1B. Figures Embedded in Analysis Docs (docs/analysis/*.md)

#	Artifact	Location	Generator	Upstream Data	Status
AN-1	perform- ance_vs_capa- city_K40.png	results/ana- lysis/capa- city_compari- son/	scripts/ana- lysis/capa- city_sensiti- vity_analysi- s.py	Optimization + simulation	✅ Reproducible
AN-2	tradeoff_dis- per- sion_rt_K40. png	results/ana- lysis/capa- city_compari- son/	scripts/ana- lysis/capa- city_sensiti- vity_analysi- s.py	Same	✅ Reproducible
AN-3	rt_heatmap_K 40.png	results/ana- lysis/capa- city_compari- son/	scripts/ana- lysis/capa- city_sensiti- vity_analysi- s.py	Same	✅ Reproducible
AN-4	full_spectru- m_summary.png	results/ana- lysis/capa- city_compari- son/	scripts/ana- lysis/capa- city_sensiti- vity_analysi- s.py	Same	✅ Reproducible
AN-5	cbd_response _comparis- on.png	results/ana- lysis/fig- ures/	notebooks/ 09_cbd_analy- sis.ipynb	CBD experi- ment data	⚠️ Notebook-only
AN-6	cbd_scenario _comparis- on.png	results/ana- lysis/fig- ures/	notebooks/ 09_cbd_analy- sis.ipynb	CBD experi- ment data	⚠️ Notebook-only
AN-7	cbd_heatmap. png	results/ana- lysis/fig- ures/	notebooks/ 09_cbd_analy- sis.ipynb	CBD experi- ment data	⚠️ Notebook-only
AN-8	queue_compar- ison_by_poli- cy.png	results/ana- lysis/fig- ures/	scripts/ana- lysis/ana- lyze_queue_m- etrics.py	Exp CSVs	⚠️ Generator has stale read path
AN-9	queue_vs_fle- et_size.png	results/ana- lysis/fig- ures/	scripts/ana- lysis/ana- lyze_queue_m- etrics.py	Same	⚠️ Same

#	Artifact	Location	Generator	Upstream Data	Status
AN-10	queue_vs_demand.png	results/analysis/figures/	scripts/analysis/analyze_queue_metrics.py	Same	⚠ Same
AN-11	queue_heatmap.png	results/analysis/figures/	scripts/analysis/analyze_queue_metrics.py	Same	⚠ Same

1C. Baseline Figures (`results/baseline/figures/`) — via `run_production_v2.py`

Artifact	Generator	Status
mean_rt_vs_K.png	run_production_v2.py	✓ Reproducible
coverage_vs_K.png	run_production_v2.py	✓ Reproducible
p95_rt_vs_K.png	run_production_v2.py	✓ Reproducible
rt_distribution_K{20,30,40}.png	run_production_v2.py	✓ Reproducible
queue_metrics_vs_K.png	run_production_v2.py	✓ Reproducible
utilization_vs_K.png	run_production_v2.py	✓ Reproducible
effect_sizes.png	run_production_v2.py	✓ Reproducible
allocation_map_K{20,30,40}.png	run_production_v2.py	✓ Reproducible
exp1_policy_comparison.png	run_production_v2.py	✓ Reproducible
exp2_fleet_sensitivity.png	run_production_v2.py	✓ Reproducible
exp3_demand_sensitivity.png	run_production_v2.py	✓ Reproducible
exp4_service_robustness.png	run_production_v2.py	✓ Reproducible
policy_comparison_panel_K20_cap2.png	run_production_v2.py	✓ Reproducible
re-sponse_time_distribution_by_policy.png	run_production_v2.py	✓ Reproducible
fleet_sensitivity_v2_dual.png	run_production_v2.py	✓ Reproducible
production_fleet_sensitivity.png	run_production_v2.py	✓ Reproducible
pub_fig{1..5}_*.png	run_production_v2.py	✓ Reproducible
project_summary_dashboard.png	run_production_v2.py	✓ Reproducible
statistical_effect_sizes.png	run_production_v2.py	✓ Reproducible
fig_*.png (EDA figures)	run_production_v2.py	✓ Reproducible

Artifact	Generator	Status
validation_*.png / verification_*.png	run_production_v2.py	✓ Reproducible
p0_vs_p2_response_time.png	run_production_v2.py	✓ Reproducible

1D. Baseline Tables (results/baseline/tables/)

Artifact	Generator	Status
descriptive_statistics.csv	run_production_v2.py	✓ Reproducible
anova_results.csv	run_production_v2.py	✓ Reproducible
posthoc_comparisons.csv	run_production_v2.py	✓ Reproducible
effect_sizes.csv	run_production_v2.py	✓ Reproducible
confidence_intervals.csv	run_production_v2.py	✓ Reproducible
queue_statistics.csv	run_production_v2.py	✓ Reproducible
table1_baseline_comparison.csv/.tex	run_production_v2.py	✓ Reproducible
table2_anova_summary.csv/.tex	run_production_v2.py	✓ Reproducible
table3_pairwise_comparisons.csv/.tex	run_production_v2.py	✓ Reproducible
table4_sensitivity_summary.csv/.tex	run_production_v2.py	✓ Reproducible
exp{1..4}*.csv	run_production_v2.py	✓ Reproducible
production_results.csv	run_production_v2.py	✓ Reproducible

1E. Analysis Figures & Tables (various results/analysis/ subdirs)

Artifact	Location	Generator	Status
Capacity comparison (21 PNGs + CSVs)	results/analysis/capacity_comparison/	capacity_sensitivity_analysis.py	✓ Reproducible
CBD focused (3 PNGs + CSVs)	results/analysis/cbd_focused_comparison/	run_cbd_focused_optimization.py	⚠ Generator writes to stale path
Distance comparison (4 PNGs + CSVs)	results/analysis/distance_comparison/	run_distance_comparison_experiment.py	✓ Reproducible
Allocation maps (3 PNGs)	results/analysis/maps/	run_optimization_comparison.py	⚠ Generator writes to stale path
Heatmaps (~160 PNGs)	results/analysis/heatmaps/	generate_all_heatmaps.py	✓ Reproducible
CBD analysis figs (6 PNGs)	results/analysis/figures/cbd_*.png	Notebook 09_cbd_analysis.ipynb + regenerate_all_figures.py	⚠ Mixed: notebook-only + stale read paths
Queue figs (5 PNGs)	results/analysis/figures/queue_*.png	analyze_queue_metrics.py	● Broken (stale read path)
Seasonal figs (3 PNGs)	results/analysis/figures/seasonal_*.png	analyze_seasonal_patterns.py	⚠ Writes to stale output path
P0 spatial figs (3 PNGs)	results/analysis/figures/p0_*.png	p0_spatial_analysis.py	⚠ Writes to stale output path
Precinct demand figs (3 PNGs)	results/analysis/figures/precinct_*.png	generate_precinct_demand_visualizations.py	● Broken (wrong column name)
Tradeoff figs (3 PNGs)	results/analysis/figures/response_time_coverage_*.png	generate_tradeoff_improved.py	⚠ Reads from stale path
Service/travel model figs (5 PNGs)	results/analysis/figures/	Notebook 04_service_travel_proxy.ipynb	⚠ Notebook-only

=== 2. ISSUES FOUND ===

Broken Reproducibility (scripts that FAIL to run)

Script	Error	Root Cause
generate_capacity_sensitivity_heatmap.py	FileNotFoundError	Reads results/capacity_comparison/ — data is at results/analysis/capacity_comparison/
analyze_queue_metrics.py	No data loaded → crash	Reads results/simulation/production/ — data is at results/analysis/simulation/production/
generate_publication_figures.py	FileNotFoundError	Reads results/simulation/production/ — same stale path
generate_summary_dashboard.py	FileNotFoundError	Reads results/tables/ and results/simulation/production/ — both stale
regenerate_all_figures.py	Partial (3 of 7 succeed)	Reads results/simulation/cbd_experiment/ and results/capacity_comparison/ — both stale
generate_precinct_demand_visualizations.py	KeyError: 'crash_rate_per_hour'	Column renamed to lambda_per_hour in demand_lambda_precinct.csv
generate_tradeoff_improved.py	Would fail	Reads results/optimization/ — data is at results/archive/optimization/

Archive Artifacts Referenced in Active (Non-Archive) Documents

Document	Artifact	Archive Path	Issue
docs/core/technical_report.md (line 478)	Fleet Sensitivity Dual	results/archive/figures/fleet_sensitivity_dual.png	Archive figure in active technical report
docs/core/technical_report.md (line 670)	Capacity Sensitivity Heatmap	results/archive/figures/capacity_sensitivity_heatmap.png	Archive figure in active report + shows “Data format issue”

Scripts Writing to Legacy Flat Paths (results/figures/ , results/tables/)

These scripts would create output in the deprecated results/figures/ or results/tables/ directories instead of the reorganized results/baseline/ or results/analysis/ structure:

Script	Stale Output Path	Correct Output Path
<code>analyze_queue_metrics.py</code>	<code>results/figures/ , results/tables/</code>	<code>results/analysis/figures/ , results/baseline/tables/</code>
<code>ana-lyze_seasonal_patterns.py</code>	<code>results/figures/ , results/tables/</code>	<code>results/analysis/figures/ , results/baseline/tables/</code>
<code>p0_spatial_analysis.py</code>	<code>results/figures/</code>	<code>results/analysis/figures/</code>
<code>generate_publication_figures.py</code>	<code>results/figures/</code>	<code>results/baseline/figures/</code>
<code>generate_summary_dashboard.py</code>	<code>results/figures/</code>	<code>results/baseline/figures/</code>
<code>regenerate_all_figures.py</code>	<code>results/figures/</code>	<code>results/baseline/figures/ or results/analysis/figures/</code>
<code>generate_tradeoff_improved.py</code>	<code>results/figures/</code>	<code>results/analysis/figures/</code>
<code>generate_precinct_demand_visualizations.py</code>	<code>results/figures/</code>	<code>results/analysis/figures/</code>
<code>run_optimization_comparison.py</code>	<code>results/optimization/ , results/maps/</code>	<code>results/archive/optimization/ , results/analysis/maps/</code>
<code>run_cbd_focused_optimization.py</code>	<code>results/cbd_focused_comparison/</code>	<code>results/analysis/cbd_focused_comparison/</code>

Stale Input Paths in Scripts

Script	Stale Read Path	Correct Read Path
<code>generate_capacity_sensitivity_heatmap.py</code>	<code>results/capacity_comparison/</code>	<code>results/analysis/capacity_comparison/</code>
<code>analyze_queue_metrics.py</code>	<code>results/simulation/production/</code>	<code>results/analysis/simulation/production/</code>
<code>generate_publication_figures.py</code>	<code>results/simulation/production/</code>	<code>results/analysis/simulation/production/</code>
<code>generate_summary_dashboard.py</code>	<code>results/tables/</code> , <code>results/simulation/production/</code>	<code>results/baseline/tables/</code> , <code>results/analysis/simulation/production/</code>
<code>regenerate_all_figures.py</code>	<code>results/simulation/cbd_experiment/</code> , <code>results/capacity_comparison/</code>	<code>results/analysis/simulation/cbd_experiment/</code> , <code>results/analysis/capacity_comparison/</code>
<code>generate_tradeoff_improved.py</code>	<code>results/optimization/</code>	<code>results/archive/optimization/</code>

Column Name Mismatch

Script	Uses	Actual Column	File
<code>generate_precinct_demand_visualizations.py</code>	<code>crash_rate_per_hour</code>	<code>lambda_per_hour</code>	<code>demand_lambda_precinct.csv</code>

Documentation Path References That Are Informational-Only Stale

The `docs/core/visualization_index.md` and `docs/core/figure_trace_guide.md` reference `results/figures/` as the location for many figures. These files have been reorganized to `results/baseline/figures/` and `results/analysis/figures/`. The index is informational (no embedded images) but is misleading.

Notebook-Only Figures (No Script Equivalent)

Figure	Notebook	Status
fig_hourly_demand.png	02_eda_spatiotemporal.ipynb	Notebook-only (also generated by run_production_v2.py)
fig_daily_demand.png	02_eda_spatiotemporal.ipynb	Same
fig_temporal_trends.png	02_eda_spatiotemporal.ipynb	Same
fig_crash_heatmap.png	02_eda_spatiotemporal.ipynb	Same
fig_precinct_density.png	02_eda_spatiotemporal.ipynb	Same
fig_firehouses_map.png	02_eda_spatiotemporal.ipynb	Same
cbd_heatmap.png	09_cbd_analysis.ipynb	Notebook-only (no script equivalent)
cbd_response_comparison.png	09_cbd_analysis.ipynb	Notebook-only
cbd_scenario_comparison.png	09_cbd_analysis.ipynb	Notebook-only
distance_matrix_heatmap.png	04_service_travel_proxy.ipynb	Notebook-only
dis- tance_matrix_coverage.png	04_service_travel_proxy.ipynb	Notebook-only
travel_time_by_tod.png	04_service_travel_proxy.ipynb	Notebook-only
tod_speed_factors.png	04_service_travel_proxy.ipynb	Notebook-only
ser- vice_time_distribution.png	04_service_travel_proxy.ipynb	Notebook-only
nhpp_arrivals_demo.png	04_service_travel_proxy.ipynb	Notebook-only

=== 3. REGENERATION RESULTS ===

Scripts That Run Successfully

Script	Output Location	Match Status	Notes
run_production_v2.py	results/baseline/	Not re-run (long-running: 810 simulations)	Paths verified correct; would produce identical structure
capacity_sensitivity_analysis.py	results/analysis/capacity_comparison/	Not re-run (requires simulation)	Paths verified correct
run_distance_comparison_experiment.py	results/analysis/distance_comparison/	Not re-run (requires simulation)	Paths verified correct
run_cbd_experiment.py	results/analysis/simulation/cbd_experiment/	Not re-run (requires simulation)	Paths verified correct
demand_modeling.py	results/baseline/figures/	Not re-run (needs raw crash data)	Paths verified correct
run_verification.py	results/baseline/simulation/verification/	Not re-run	Paths verified correct
run_validation_pilots.py	results/baseline/simulation/validation_pilot/	Not re-run	Paths verified correct
p0_spatial_analysis.py	⚠️ results/figures/ (stale)	✅ Ran successfully	Output went to wrong dir; should write to results/analysis/figures/

Scripts That FAIL

Script	Error	Attempted Fix	Notes
<code>generate_capacity_sensitivity_heatmap.py</code>	FileNotFoundError	Path fix needed	Would need <code>results/capacity_comparison/</code> → <code>results/analysis/capacity_comparison/</code>
<code>analyze_queue_metrics.py</code>	No data found	Path fix needed	Would need <code>results/simulation/production/</code> → <code>results/analysis/simulation/production/</code>
<code>generate_publication_figures.py</code>	FileNotFoundError	Path fix needed	Same stale simulation path
<code>generate_summary_dashboard.py</code>	FileNotFoundError	Two path fixes needed	Both <code>results/tables/</code> and <code>results/simulation/production/</code> are stale
<code>regenerate_all_figures.py</code>	Partial success (3/7)	Two path fixes needed	CBD + capacity paths stale
<code>generate_precinct_demand_visualizations.py</code>	KeyError	Column rename fix	<code>crash_rate_per_hour</code> → <code>lambda_per_hour</code>

=== 4. RECOMMENDED ACTIONS ===

Keep As-Is (No Changes Needed)

These artifacts and their generators are correctly configured:

1. **All** `results/baseline/` **artifacts** — Generated by `run_production_v2.py` with correct paths
2. `results/analysis/capacity_comparison/` **artifacts** — `capacity_sensitivity_analysis.py` paths are correct
3. `results/analysis/distance_comparison/` **artifacts** — `run_distance_comparison_experiment.py` paths correct
4. `results/analysis/simulation/` **artifacts** — Experiment runners have correct paths
5. `demand_modeling.py` — Writes to `results/baseline/figures/` correctly
6. `run_verification.py` and `run_validation_pilots.py` — Both correct

Fix Script Input/Output Paths (13 scripts)

Script	Fix Required
<code>analyze_queue_metrics.py</code>	Read: <code>results/simulation/production/</code> → <code>results/analysis/simulation/production/</code> ; Write figures: → <code>results/analysis/figures/</code> ; Write tables: → <code>results/baseline/tables/</code>
<code>analyze_seasonal_patterns.py</code>	Write figures: → <code>results/analysis/figures/</code> ; Write tables: → <code>results/baseline/tables/</code>
<code>generate_capacity_sensitivity_heatmap.py</code>	Read: <code>results/capacity_comparison/</code> → <code>results/analysis/capacity_comparison/</code> ; Write: → <code>results/analysis/figures/</code>
<code>generate_publication_figures.py</code>	Read: <code>results/simulation/production/</code> → <code>results/analysis/simulation/production/</code> ; Write: → <code>results/baseline/figures/</code> ; Fix CAPACITY=5 → CAPACITY=2
<code>generate_summary_dashboard.py</code>	Read tables: <code>results/tables/</code> → <code>results/baseline/tables/</code> ; Read sim: <code>results/simulation/production/</code> → <code>results/analysis/simulation/production/</code> ; Write: → <code>results/baseline/figures/</code>
<code>regenerate_all_figures.py</code>	Read CBD: <code>results/simulation/cbd_experiment/</code> → <code>results/analysis/simulation/cbd_experiment/</code> ; Read capacity: <code>results/capacity_comparison/</code> → <code>results/analysis/capacity_comparison/</code> ; Write: split between <code>results/baseline/figures/</code> and <code>results/analysis/figures/</code>
<code>generate_tradeoff_improved.py</code>	Read: <code>results/optimization/</code> → <code>results/archive/optimization/</code> ; Write: → <code>results/analysis/figures/</code> ; Fix CAPACITY=5 → document legacy
<code>generate_precinct_demand_visualizations.py</code>	Fix column: <code>crash_rate_per_hour</code> → <code>lambda_per_hour</code> ; Write: → <code>results/analysis/figures/</code>
<code>p0_spatial_analysis.py</code>	Write: <code>results/figures/</code> → <code>results/analysis/figures/</code>
<code>run_optimization_comparison.py</code>	Write results: <code>results/optimization/</code> → <code>results/archive/optimization/</code> ; Write maps:

Script	Fix Required
	<code>results/maps/ → results/analysis/maps/ ;</code> Write figures: <code>→ results/baseline/figures/</code>
<code>run_cbd_focused_optimization.py</code>	Write: <code>results/cbd_focused_comparison/ →</code> <code>results/analysis/cbd_focused_comparison/</code>

Regenerate and Replace

After path fixes, these artifacts should be regenerated:

1. **capacity_sensitivity_heatmap.png** — Currently in archive and shows “Data format issue” per user screenshot. After fixing `generate_capacity_sensitivity_heatmap.py`, regenerate to `results/analysis/figures/`
2. **Queue analysis figures** — After fixing `analyze_queue_metrics.py` paths, regenerate to `results/analysis/figures/`
3. **Publication figures** — After fixing `generate_publication_figures.py`, regenerate to confirm match with `results/baseline/figures/` versions

Move to Archive

Artifact	Current Location	Reason
No artifacts need moving	—	Current archive contents are already correctly placed

Fix Technical Report References

Line	Current Reference	Recommended Fix
478	<code>results/archive/figures/fleet_sensitivity_dual.png</code>	Replace with <code>results/baseline/figures/fleet_sensitivity_v2_dual.png</code> (the current cap=2 version)
670	<code>results/archive/figures/capacity_sensitivity_heatmap.png</code>	Regenerate to <code>results/analysis/figures/capacity_sensitivity_heatmap.png</code> and update reference

Update Documentation Indexes

Document	Issue
docs/core/visualization_index.md	~100+ references to <code>results/figures/</code> should be updated to <code>results/baseline/figures/</code> or <code>results/analysis/figures/</code> as appropriate
docs/core/figure_trace_guide.md	Same — all figure locations reference <code>results/figures/</code> which is now a stub directory
docs/core/data_usage_guide.md	References <code>results/figures/*.png</code> generically; should be updated to the reorganized paths

Summary Statistics

Category	Count
Total artifacts audited	~120 figures + ~40 tables
Fully reproducible (script + correct paths)	~75 figures, ~30 tables
Broken scripts (won't run)	6 scripts
Scripts with stale output paths	10 scripts
Scripts with stale input paths	6 scripts
Archive artifacts in active docs	2 (technical report)
Notebook-only figures (no script)	15 figures
Column name mismatches	1 (<code>crash_rate_per_hour</code> vs <code>lambda_per_hour</code>)

Root Cause

All issues stem from **a single organizational change**: the directory reorganization from flat `results/figures/`, `results/tables/`, `results/simulation/`, `results/optimization/`, `results/capacity_comparison/` into the tiered `results/baseline/`, `results/analysis/`, `results/archive/` structure. **The scripts were not updated to match the new directory structure.** The data and figures themselves are valid; only the paths in the generating scripts are stale.

Priority Ranking

1. ● **P0 — Fix the 6 broken scripts** (can't regenerate at all)
2. ● **P1 — Fix technical report archive refs** (TR lines 478, 670)

3.  **P2 — Fix 10 scripts with stale output paths** (would write to wrong dirs)
4.  **P3 — Update visualization_index.md and figure_trace_guide.md** (informational)
5.  **P4 — Create script equivalents for notebook-only figures** (nice-to-have)